

IMPORTANT UPDATE!

The forum that was previously hosted on Wix is now being moved to Github. We will maintain this for future information only and posting or editing is being disabled.

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Liam Pulido
Dec 28, 2024

Edited: May 06 ⋮

Setting Up On A CNC Shield v4.

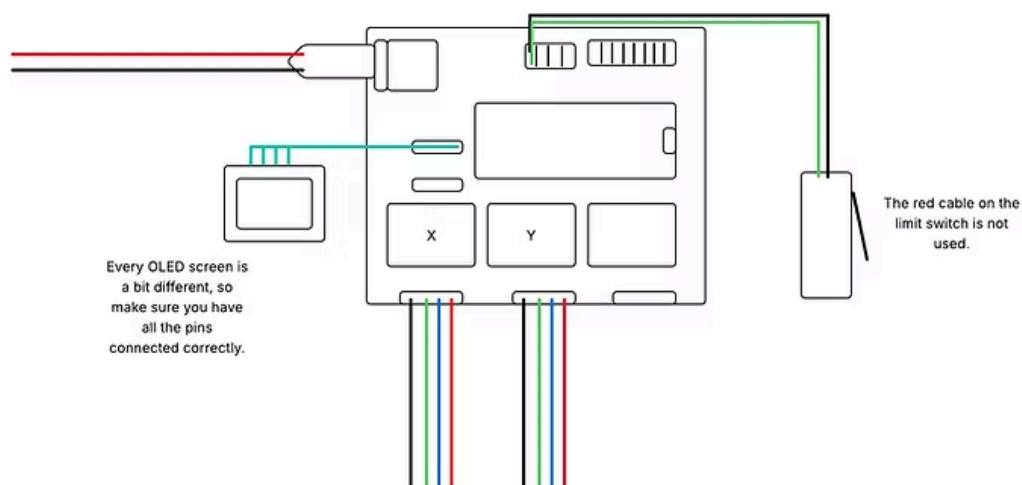
in [Support](#)

This is a thread about using the CNC Shield v4 as the main board. If you plan to do this, buy this kit:

https://www.aliexpress.us/item/2251832035811475.html?spm=a2g0o.detail.pcDetailTopMoreOtherSeller.10.3fe9i0iWi0iW2v&gps-id=pcDetailTopMoreOtherSeller&scm=1007.40196.366991.0&scm_id=1007.40196.366991.0&scm-url=1007.40196.366991.0&pvid=5c83e78e-97fb-48de-a5c7-35db0da5b120&t=gps-id:pcDetailTopMoreOtherSeller.scm-url:1007.40196.366991.0.pvid:5c83e78e-97fb-48de-a5c7-35db0da5b120.tpp_buckets:668%232846%238111%231996&pdp_ext_f=%7B%22order%22%3A%2210%22%2C%22eval%22%3A%221%22%2C%22sceneld%22%3A%2230050%22%7D&pdp_npi=4%40dis%21USD%2119.99%2119.99%21%21%2119.99%2119.99%21%402103010e17397411430922198e1202%2155305829959%21rec%21US%216242698492%21X&utparam-url=scene%3ApcDetailTopMoreOtherSeller%7Cquery_from%3A This may be more expensive then the \$2

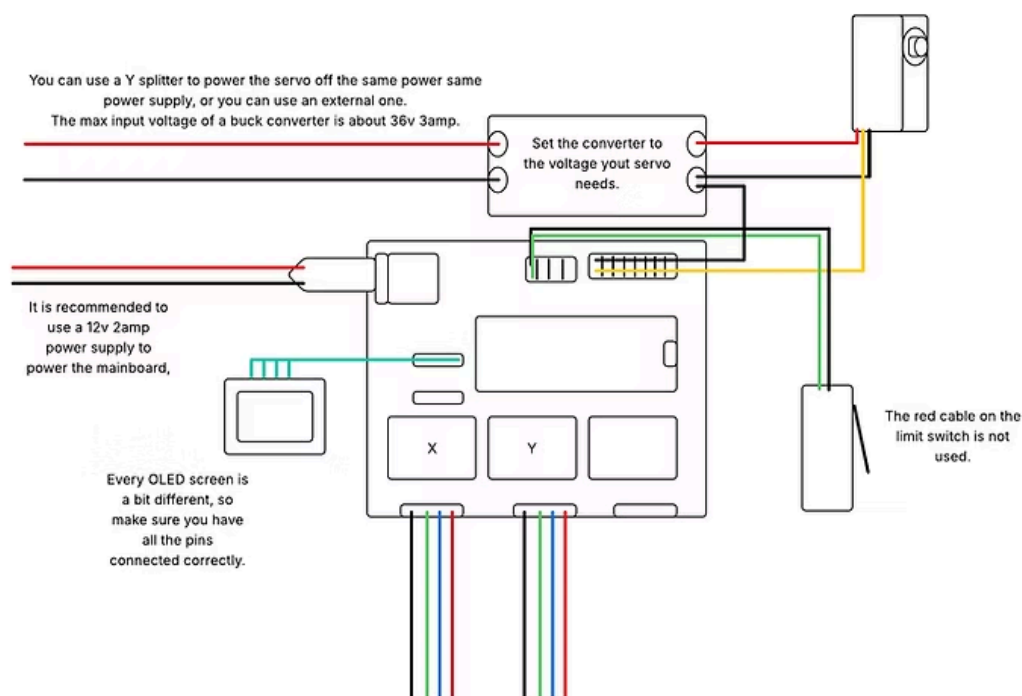
"cloned" boards, but it is entirely worth it. The cloned boards have many problems that require cutting PCB and soldering to fix. To find out more about this, look up "Fix Cloned CNC Shield" and an instructable should show up. The expensive Keyestudio kits work right out of the box, and come with the required stepper drivers. If you need any help, just ask.

Here is the pinout diagram for the CNC Shield:



Chameleon Pinout for the CNC Shield v4

Here is the pinout for adding a buck converter powered servo:



Chameleon Pinout for the CNC Shield v4, with an externally powered servo

I will probably not update this thread much (or whatever the posts will look like on the github) because I am busy developing my own budget MMU (for klipper printers only). That's it for now!
Liam

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4

194 Comments



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Jim Kramer
Dec 29, 2024



✓ Best Answer

I have it working (in breadboard mode) with a CNC V4. Pins for my V4 are in the image [below](#). Micro steps will depend on jumpers on the V4. Had an issue with the steppers and how they are wired with the flat cable that came with them. Had to swap some pins. Easy to check what pairs to use by shorting stepper pins together to find the ones that cause the stepper to 'turn hard' by hand. direction can be changed by swapping the connector on the board around.

```
SelectorFirmwareMk4-nano.ino  SparkFunSX1509.h
140  SX1509 io;                  // Create an SX1509 object to be used thr
141
142
143  // defines pins numbers - 3D Chameleon Board
144  #define extEnable 8
145  #define extStep 2
146  #define extDir 5
147
148  #define selEnable 8
149  #define selStep 3
150  #define selDir 6
151
152  #define trigger A3
153  #define s_limit A4
154  #define filament A5
155
156  const int counterclockwise = HIGH;
157  const int clockwise = !counterclockwise;
158
159  const int stepsPerRev = 200;
160  const int microSteps = 16;
161  const int speedDelay = 170;
162
163  const int defaultBackoff = 10;
164
```

No on GRBL. You will overwrite what might be on the nano.
the code above has the switch connected to A3 (and ground) I had a 3 wire patch cable connected to the switch and use 2 of the connectors to connect to the V4.
be aware V4 boards are not all the same. I ended up with this one
Free Shipping! New! Keystudio Cnc Shield V4 +3pcs A4988 Driver+ Nano Ch340 For Arduino Cnc
- Integrated Circuits - AliExpress
I'll get some pics later.

Show more replies

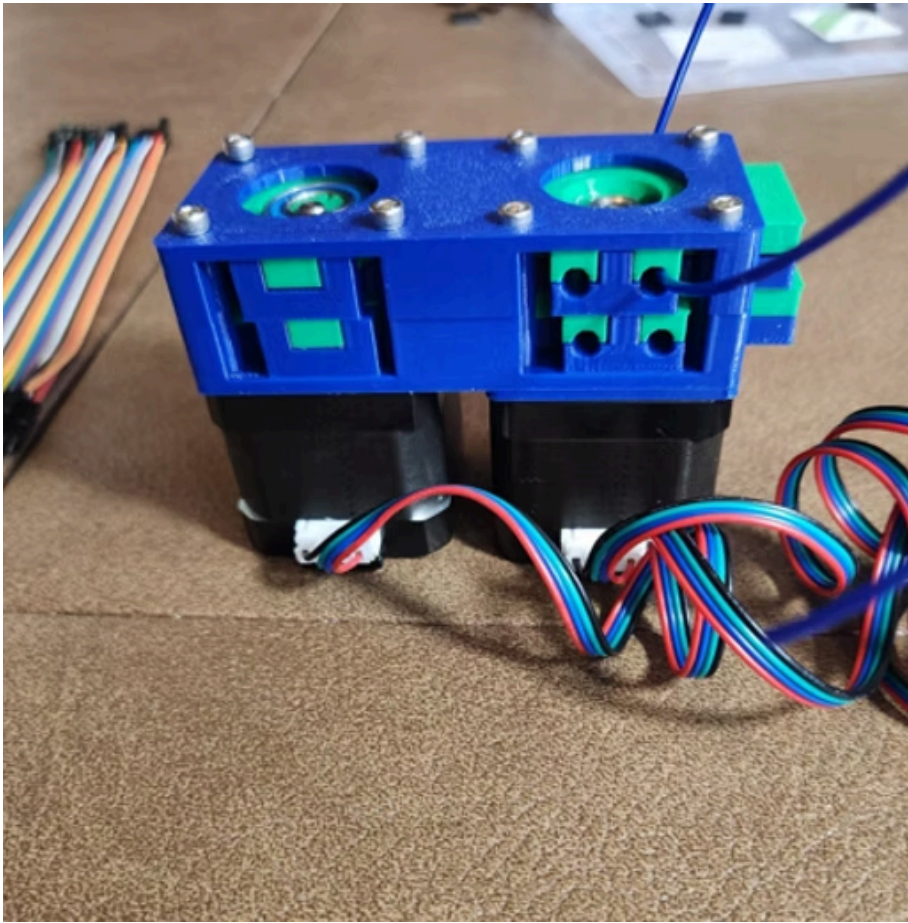


Jim Kramer
Dec 29, 2024

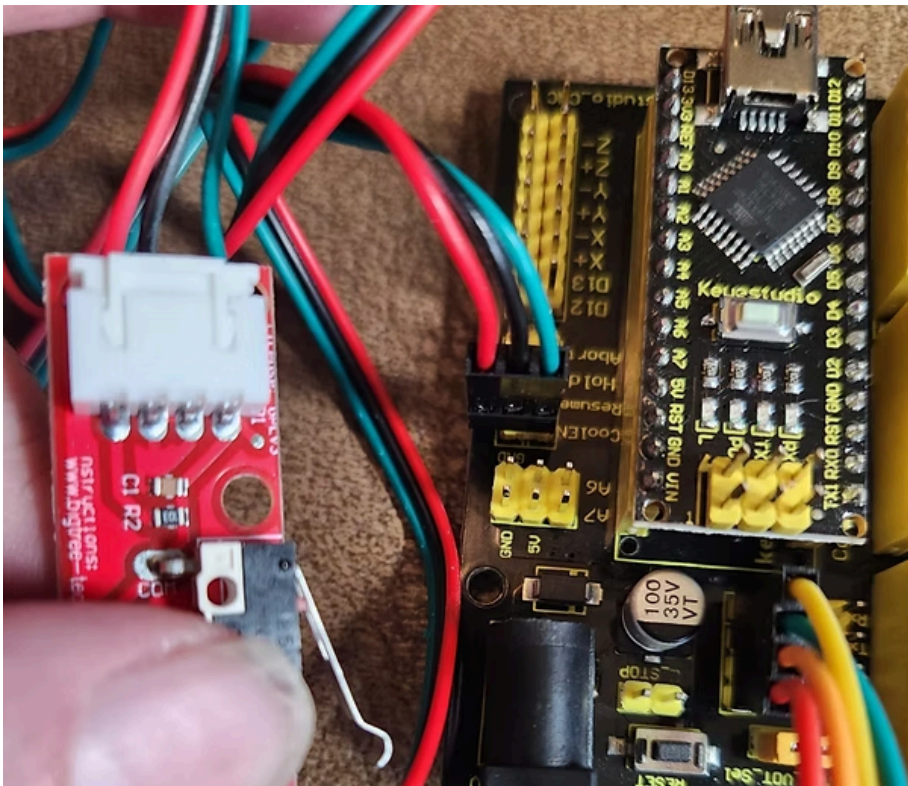


✓ Best Answer

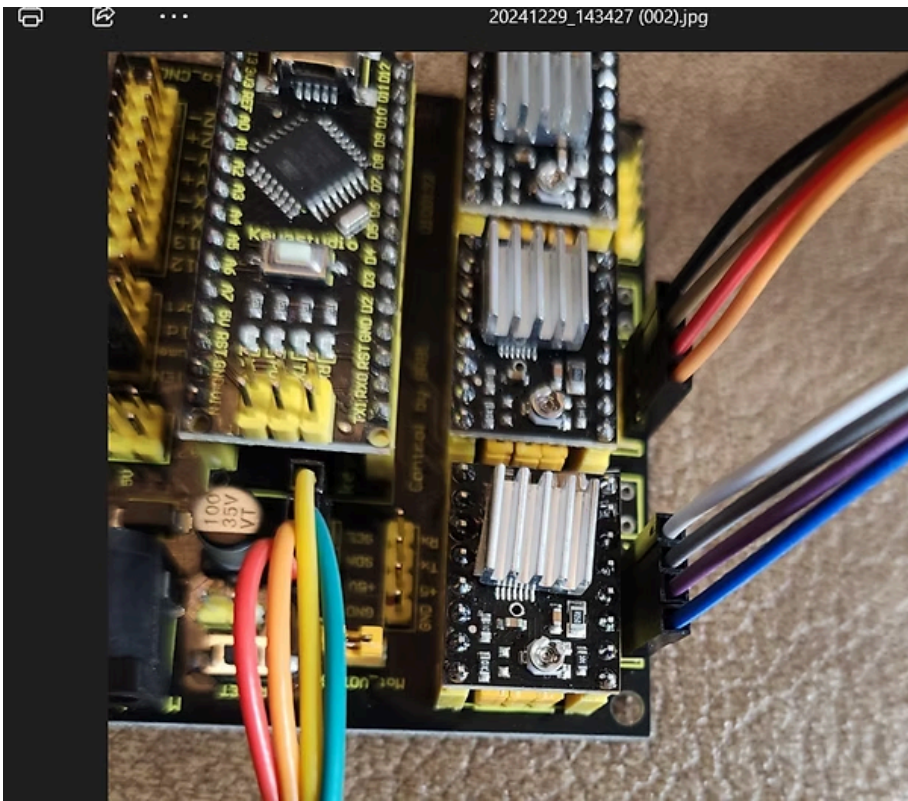
Replying to [Jim Kramer](#)
adding some pics.



switch and connections



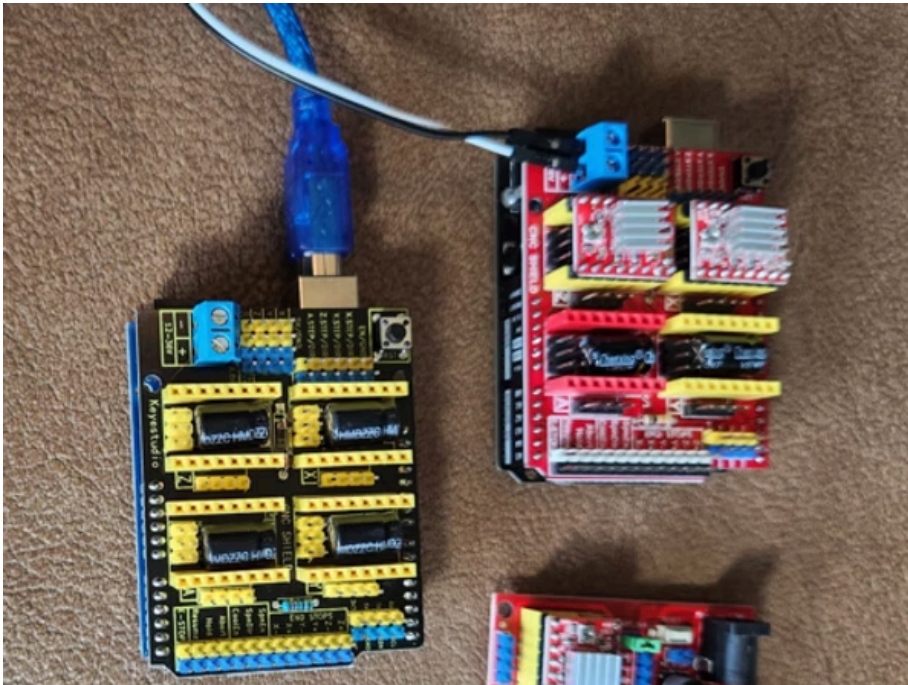
steppers and display connections



display



other setups



Jim Kramer
Dec 29, 2024



✓ Best Answer

Replying to [tazarian44](#)

yes, I got it to work. Not all V4 boards are the same so that is why I got the Keystudio one. Here are some links that you can look at also. [How to Use the CNC V4 Board \(despite Its "quirks"\) : 4 Steps - Instructables](#)

[Fix Cloned Arduino NANO CNC Shield : 10 Steps - Instructables](#)

[Worst CNC shield 12v /24 volt board GRBL v4 Arduino Nano \(Problems, fix, Mods, flaws, corrections\)](#)

did you short the stepper pins to find the pairs?



Jim Kramer
Dec 30, 2024



✓ Best Answer

Replying to [tazarian44](#)

when I was having trouble with my CNC boards, I used this code to test. Not my code but simple to use for testing.

```
#define EN      8    // stepper motor enable, low level effective
#define X_DIR   5    //X axis, stepper motor direction control
#define Y_DIR   6    //y axis, stepper motor direction control
#define Z_DIR   7    //z axis, stepper motor direction control
```

```

#define X_STP 2 //x axis, stepper motor control
#define Y_STP 3 //y axis, stepper motor control
#define Z_STP 4 //z axis, stepper motor control
/*
// Function: step -control the direction and number of steps of the stepper motor
// Parameter: dir -direction control, dirPin corresponds to DIR pin, stepperPin corresponds
to

step pin, steps is the number of steps.
// no return value
*/
void step(boolean dir, byte dirPin, byte stepperPin, int steps)
{
    digitalWrite(dirPin, dir);
    delay(50);
    for (int i = 0; i < steps; i++) {
        digitalWrite(stepperPin, HIGH);
        delayMicroseconds(800);
        digitalWrite(stepperPin, LOW);
        delayMicroseconds(800);
    }
}

void setup(){// set the IO pins for the stepper motors as output
    pinMode(X_DIR, OUTPUT); pinMode(X_STP, OUTPUT);
    pinMode(Y_DIR, OUTPUT); pinMode(Y_STP, OUTPUT);
    pinMode(Z_DIR, OUTPUT); pinMode(Z_STP, OUTPUT);
    pinMode(EN, OUTPUT);
    digitalWrite(EN, LOW);
}

void loop(){
    step(false, X_DIR, X_STP, 3200); // x axis motor rotates CCW for 1 circle, as in 200 steps
    step(false, Y_DIR, Y_STP, 3200); // y axis motor rotates CCW for 1 circle, as in 200 steps
    step(false, Z_DIR, Z_STP, 3200); // z axis motor rotates CCW for 1 circle, as in 200 steps
    delay(1000);
    step(true, X_DIR, X_STP, 3200); // X axis motor rotates CW for 1 circle, as in 200 steps
    step(true, Y_DIR, Y_STP, 3200); // y axis motor rotates CW for 1 circle, as in 200 steps
    step(true, Z_DIR, Z_STP, 3200); // z axis motor rotates CW for 1 circle, as in 200 steps
    delay(1000);
}

```



Kendall Leake
May 28



Hey Liam. I have been staring at this thread off and on for a while now. Lots of information! I hope you are still around even after the Github move.

I am trying to test Keystudio board after programming the board. I have the steppers, the Nano board and the switch attached to the Keystudio and the light on the Nano lights then slowly fades, is that normal? I press and hold the switch and nothing happens.



Catz 4 Catz
May 16



Hello Liam,

Long time please let me know how your MMU (for kilpper go's) as I have converted my 3 ender 5's to klipper and would be very interested. As MARLIN is a BIG PAIN IN MY ASS 🙄🙄🙄🙄
Thank You



Liam Pulido
May 18



Replying to [Catz 4 Catz](#)

I decided to make a 3MS (<https://3ms.3dcoded.xyz/setup/bom/>). I do agree that marlin sucks. I am NEVER going back to that. I will note that it can be kind of hard to flash klipper to the Arduino, do to bugs with klipper, but I have successfully gotten it working. Another note: you can still use the chameleon hardware, but remove the need for a limit switch. I have read up on that, but have not done it myself. If you do end up using the chameleon, I would recommend going to the happy-hare repo (<https://github.com/moggieuk/Happy-Hare>) then going to the wiki and looking at the 3Dchameleon quickstart. Happy hare does require a filament runout switch, but I assume that you already have one. If you do not, you can easily find a model that converts your chameleon limit switch into a runout switch.

Here are a few helpful links:



GitHub -
pannonbeard/kl
Trying to make i
an arduino, cnc
Klinper

github.com

GitHub - pannonbeard/klipper_multi_color: Trying to make m...
Trying to make my own AMS with an arduino, cnc sheild and Klipper - ...



Firmware no
longer builds fo
Atmega 328p ·
Issue #4938 ·
Klinper3d/klipn

github.com

Firmware no longer builds for Atmega 328p · Issue #4938 · Kl...
Looks like the image is too big for flash. Not sure what commit exactly ...

You may want to ask more on the 3MS/happy hare pages since they know more about klipper setups than I do.

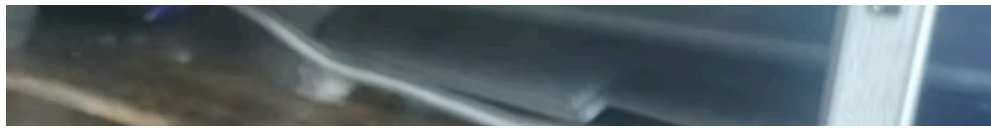


Ліза Москаленко
May 09



Hello everyone. I have a question - where in the firmware can I set a pause or delay for slower scrolling of the menu?





Bill 
May 09



Replying to [Ліза Москаленко](#)

There are a few places you can play with... just look for the Delay methods... they take a millisecond parameter.

Bill



Ace6879
Apr 30



What's the wiring for the buck converter, the shield and the servo look like?

[Show more replies](#)



Ace6879
May 13



Replying to [Liam Pulido](#)

I had actually bought a Bambu 4 way and it works perfectly. The one I had eventually gave way and snapped in half. I haven't touched the start G-Code, but thought about it. I'll give the video a go then!



Liam Pulido
May 13



Replying to [Ace6879](#)

Good choice. Bambu has the best Y adapter since it's injection moulded and does not have layer lines that filament can catch on. You are definitely going to finish your MMU sooner than me. I hope you can have some fun with it after everything is complete.



Liam Pulido
May 16



Replying to [Liam Pulido](#)

I do not know if this even went through, but I am making a new github discussions post for this.



Kasen Meyerhoffer
Apr 29



Hey Liam, I have everything working besides the chameleon after 2 or 3 color changes. The first one gets the color correct, even if it is any color. But when it switches to a color it has a chance to

move 2 colors and that isn't good. And then on the third color switch it always moves 2 colors. When I home my chameleon it like resets it and so it does it again, it gets the first color right and then the rest are wrong. Thanks for the help.



Liam Pulido
Apr 29



Replying to [Kasen Meyerhoffer](#)

Great that you got it working, not so great that it isn't working. Do you have Klipper set up with your printer? If you do, I think directly configured klipper macros may fix your problem, since it does not rely on the pre-flashed firmware or button pressing. I have not tested it, but since Klipper stores your config in a text file, I think it would be easier to modify. The color selection problem is why I moved over to a 3MS-based system (I am actually making my own, I will post on the forum when it is done) that has individual extruders.

Edited



Kasen Meyerhoffer
Apr 29



Replying to [Liam Pulido](#)

I do not have a Klipper, do you think it would be worth it to invest in one for my 3d printer?



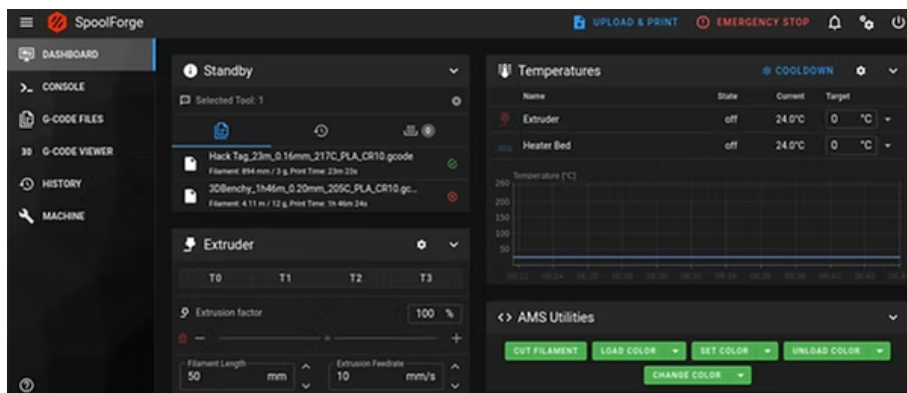
Liam Pulido
Apr 30



Replying to [Kasen Meyerhoffer](#)

It depends if your printer is even compatible with klipper firmware. Which printer do you have?

In my opinion, changing to klipper is worth every cent. you can do it for ~\$25 and it is so worth it. It gives you a web interface, the ability to print wirelessly (that's not new for printers these days, though), and so much more. Here's a pic of what one of the web interface options look like (and what my custom MMU controls look like):



There is no camera, yet, but I am going to get one and make a mount for it.

 juanjmr
Apr 23

⋮

Hi Jim! I'm also building my project with a VShiel v4 CNC board, with the tracks arranged, but I have several questions. Could you please help me? I have three questions. First, for the Drv8825 controllers, do you know if the three jumpers need to be set so the motors operate at 1/16, as seems appropriate? Second, where on the board should the microswitch be placed? And third, what is the correct final code to upload to Arduino with this board? Thanks!!

 Liam Pulido
Apr 29

⋮

Replying to [juanjmr](#)

Place the microswitch as shown:

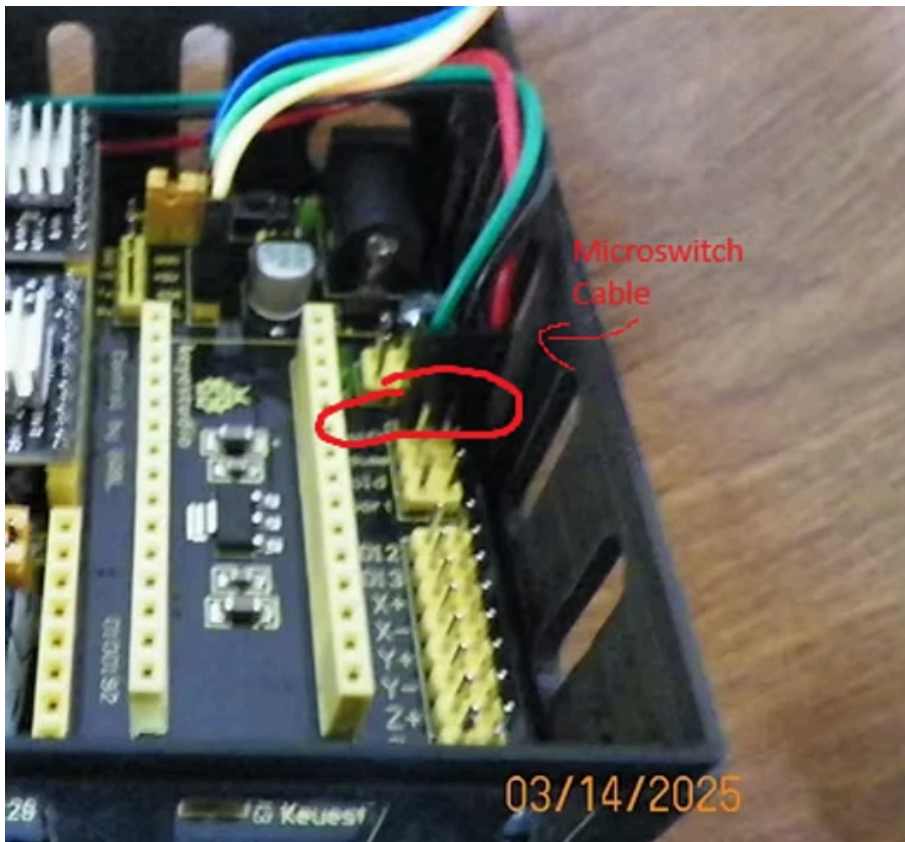


Photo taken by Catz 4 Catz (just giving proper credit)

If you still need clarification, send me a pic of your current pinout and I can help you from there.

 Ace6879
Apr 20

⋮

Btw, my issue seems to have just been power supply. It almost worked with a 2A but I decided to experiment and bought a 5A and it made a big difference.

 Liam Pulido
Apr 27

⋮

Replying to [Ace6879](#)

Hmm. 2 amp should be enough. Try adjusting the screws on the drivers to allow more current. I said to lower the current earlier, and that may be the problem.

Sorry for taking so long. I just got back from mexico :)



Ace6879
Mar 31



I bought another CNC shield thinking that was the issue but it didn't make a difference. I ended up burning thru 5 drivers. Is it possible that the drivers can burn out if there is too much resistance from a motor that gets jammed?

[Show more replies](#)



Liam Pulido
Apr 06



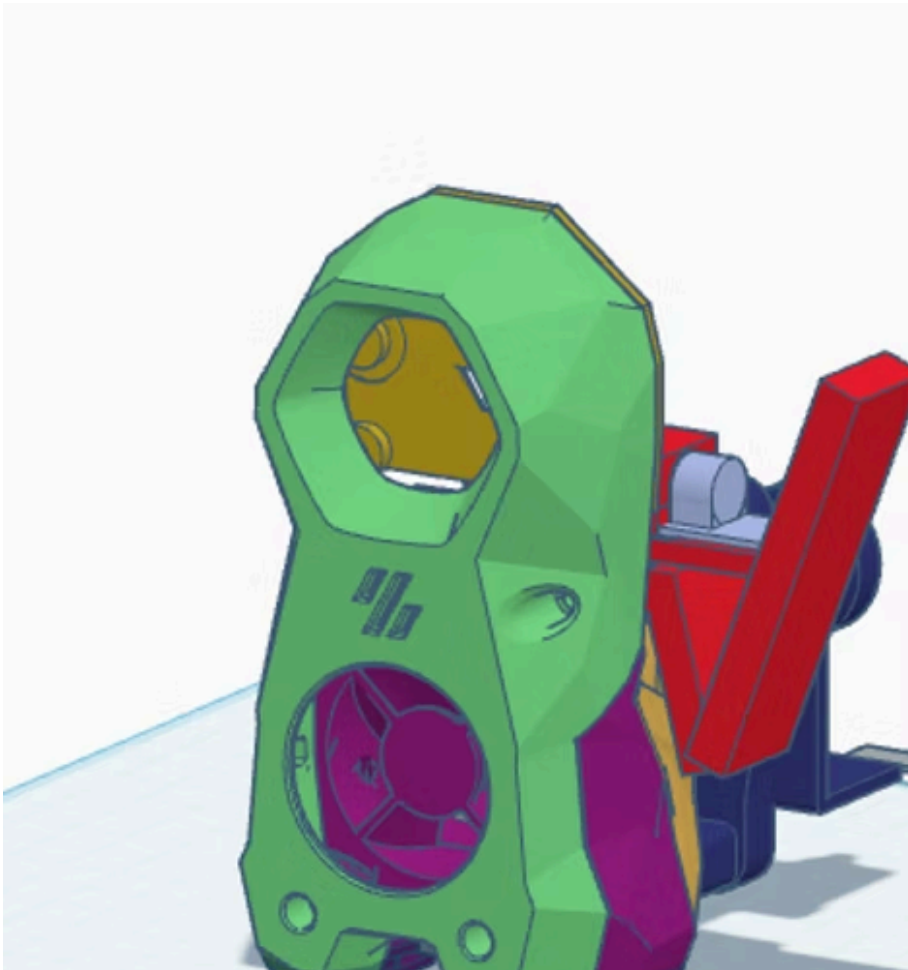
Replying to [Ace6879](#)

Also, if you plan to keep your ender on a Bowden setup, then you can probably use my (still being designed) Stealthburner fork, the Color Burner. It's a modified version that I have been remixing to use mostly stock hardware (stock hotend, uses the stock hotend cooling fan, and mounts to the OG Extruder carriage) and have a cheap filament cutter built in. You only need a bit of external hardware.

Current BOM (may change):

- Voron Stealthburner (just the 5015 fan, heat inserts, bolts, washers, and screws. None of the fancy lighting and sensor stuff is needed)
- No. 17 X-acto blade
- 9x24mm Spring (the one you would find in the ender-3 Bowden extruder, so if you upgrade to the M4, just take it from there)
- M3 Heat Inserts
- BL/3D touch (optional, but I have heard it is worthwhile)
- Color Burner modified parts (will release when they're done)
- m2 screw kit (mainly 20mm sizes)
- PC4-M10 Pneumatic Fitting

That's about it, but it is still in development, so it may change. Also, it's being designed for the CR-10 (pretty much just a giant ender 3) so it most likely fits the ender 3, but I will not be able to confirm. Another note, I will not be able to respond from the 20th through the 28th, so just a heads up.



Joseph E
May 11



Replying to [Liam Pulido](#)

Have you actually made one of these, or is it still just a concept?



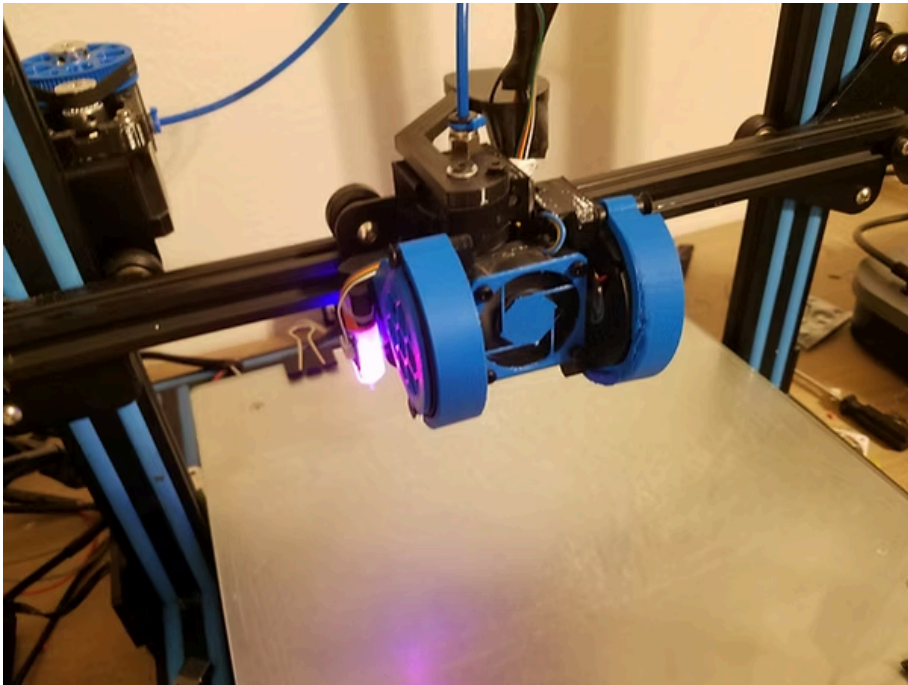
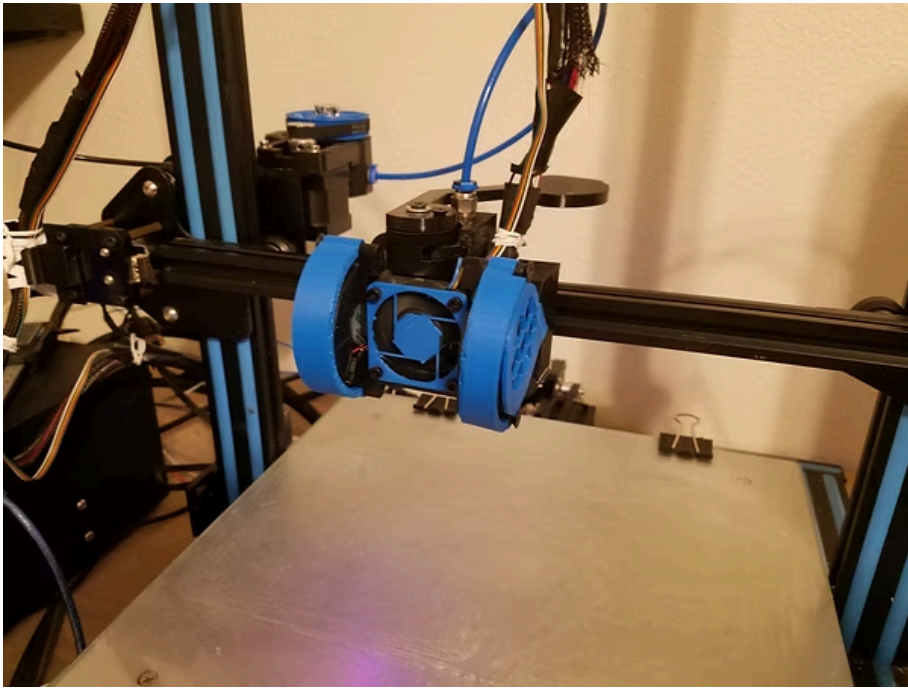
Liam Pulido
May 11



Replying to [Joseph E](#)

I did make a toolhead, but I wanted dual 5015 cooling rather than just the single fan that the SB has, so I made a new design and it is working great!

Pics:



It cuts perfectly!

BOM:

2 5015 fans

M4x20 screws

PFTG or better



Kasen Meyerhoffer
Mar 26



I have a couple of questions, first where do I place the 3d clippy servo on the CNC Shield v4 and what orientation do I do for it. Another question is what size of servo should I use for a modified 3d

clippy, <https://www.printables.com/model/778707-3dclippy-generic-filament-cutter> I personally have modified it for my printer and circumstances. My last question is my Cam broke and I printed a new one and now every time I select one T0 it pulls T1. I only have 3 filaments so I don't know if T2 and T3 are also doing that. I have modified the code to help a little but it still does it. Thanks.

Show more replies



Liam Pulido
Mar 28



Replying to [Kasen Meyerhoffer](#)

Oh, make sure to reprint everything on the same print in PETG with .2 mm layer height and 100% infill. This is the kind of servo Bill recommends: https://www.amazon.com/Torque-Motors-Waterproof-Steering-Control/dp/B0DDKN9RS6/ref=sr_d_dpds_fsdp4star_fa_xcat_cheapdynam_d_sccl_2_4/135-4448785-9319159?pd_rd_w=K5THO&content-id=amzn1.sym.189ba0e8-243a-408f-947d-77c5ac846d2e&pf_rd_p=189ba0e8-243a-408f-947d-77c5ac846d2e&pf_rd_r=8DKW5ZRK9SP2JSWR0323&pd_rd_wg=s3ftv&pd_rd_r=ba40394c-0911-475e-b176-9241d4adb27e&pd_rd_i=B073F92G2S&th=1

Choose the 20 or 25 kg (preferably 25) with 180 degrees.

I do not know if the one you linked will work (or not work) so you choose that at your own discretion. As for a buck booster, you will need:

Buck Booster: <https://www.amazon.com/dp/B081N6WWJS>

Power supply to wire adapter: <https://www.amazon.com/Qaoquda-Connector-Terminal-Headphone-Converter/dp/B07JMY5XXT>

Power supply splitter: <https://www.amazon.com/Splitter-Surveillance-Adapter-Y-Cable-Security/dp/B082Y1PQ6C>

You will use this to split a wire of the 12v power supply, then convert the splitter to a wire terminal so you can connect it to the buck booster (which you will need to set to your servos highest voltage) which then can connect to your servo.

You will need to be careful while doing this, as I accidentally fried a cloned Arduino while attempting (due to user error). Side note for the servo: Depending on what your budget is, you could get both and see if they both work the same. My budget is very low, so I pretty much only bought the necessary components.

Edited

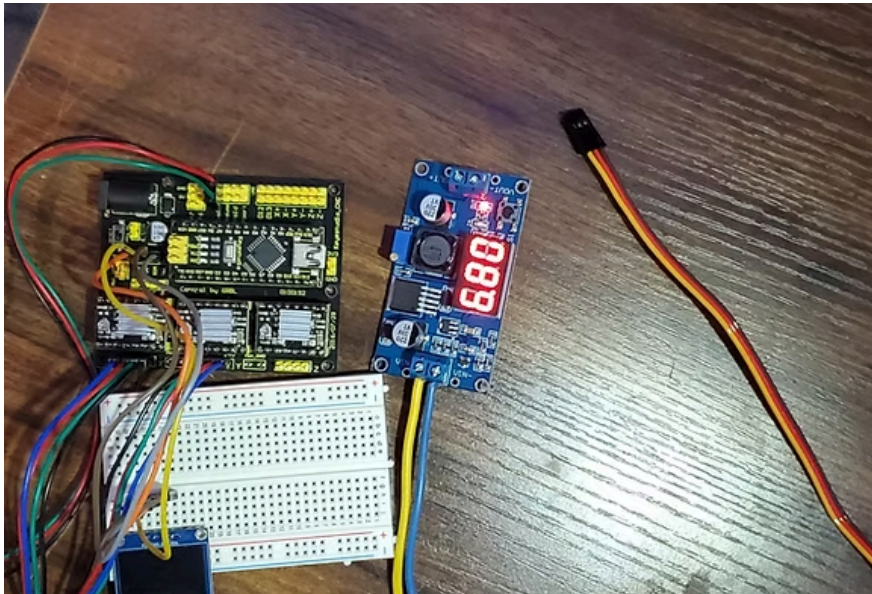


Kasen Meyerhoffer
Mar 30



Replying to [Liam Pulido](#)

So I have everything now and my buck booster is set to 6.8 volts(what my servo is rated for). What wires get plugged into the buck booster and what wires go into my board from the servo? also what port does the wires go into my board?



Liam Pulido
Apr 30



Replying to [Kasen Meyerhoffer](#)

I never got back to you on this, but I made the answer on this comment:

<https://www.3dchameleon.com/forum/main/comment/d9adf8ff-c911-4253-848c-575d6a620060?postId=6770a14a797c5c8b5ad99781>



Catz 4 Catz
Mar 22



Hello Liam,

I remember a wise man once told me NEVER PLUG POWER IN WITH THE STEPPERS MOTORS UNPLUG if you do the drivers/board may SMOKE ! Can't rember who that was Oh that was YOU ! LOL 🐱🐱



Liam Pulido
Mar 22



Replying to [Catz 4 Catz](#)

It probably wasn't that, because then both drivers would be fried, not just one. I think it could be a faulty driver or incorrect cable/driver placement.



Catz 4 Catz
Mar 22



Hello Liam,

Sounds to me like he fried something as the only problem I had was the BAD TEST CODE everything worked rite out of the box

Thank You 🐱🐱



Liam Pulido
Mar 22



Replying to [Catz 4 Catz](#)

I think that he may have a bad stepper driver. The kit should have come with 3, so he can replace the (maybe) fried driver with the spare.



Catz 4 Catz
Mar 22



Hello Liam,

Didn't mean to step on your toes just tried to help sounds like you know this person so I will let you deal with it Thanks again for all your help with my problems.

Thank You 🐱🐱



Liam Pulido
Mar 22



Replying to [Catz 4 Catz](#)

It's fine. Any help is appreciated! I would like to help him because I sold my chameleon unit to him, so I should help get his unit to work. After all, who wants to waste \$155?



Ace6879
Mar 21



I swapped around cables and motors. I'll try and send a video when I work on it again



Liam Pulido
Mar 22



Replying to [Ace6879](#)

Since this problem isn't directly related to the chameleon, it may be worth posting about this on the Arduino forums. I have never seen this problem before, so I will have to do some more research into this.



Catz 4 Catz
Mar 21



Hello Ace6879,

On the one that works try swaping your stepper motor cables and see if the one that works still works. Sometimes stepper cables are not as they seem and are configured wrong.

Thank You



Catz 4 Catz
Mar 21



Hello Ace6879,

Can you send a picture of your setup (board) ? If your using the keystudio board you don't need to move any jumpers it just works out of the box.

Ace6879
Mar 21

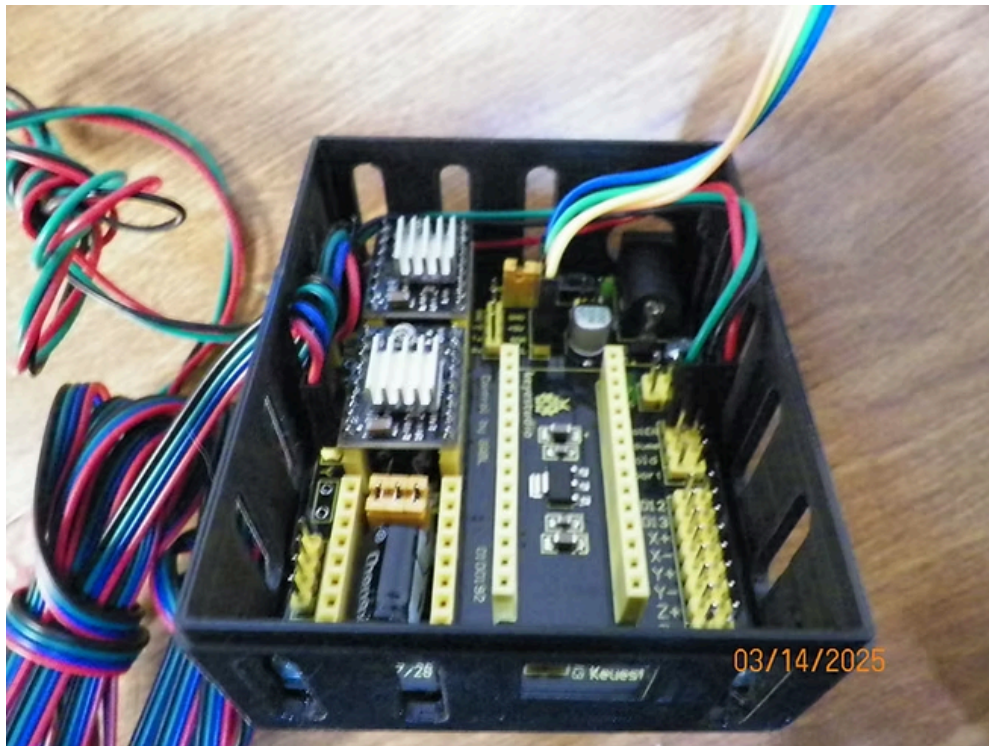


Yes I switched drivers around and same issue. I tested each motor individually so I don't think they are the issue and all the drivers work. What about the jumpers under the drivers? I'm not sure what they do

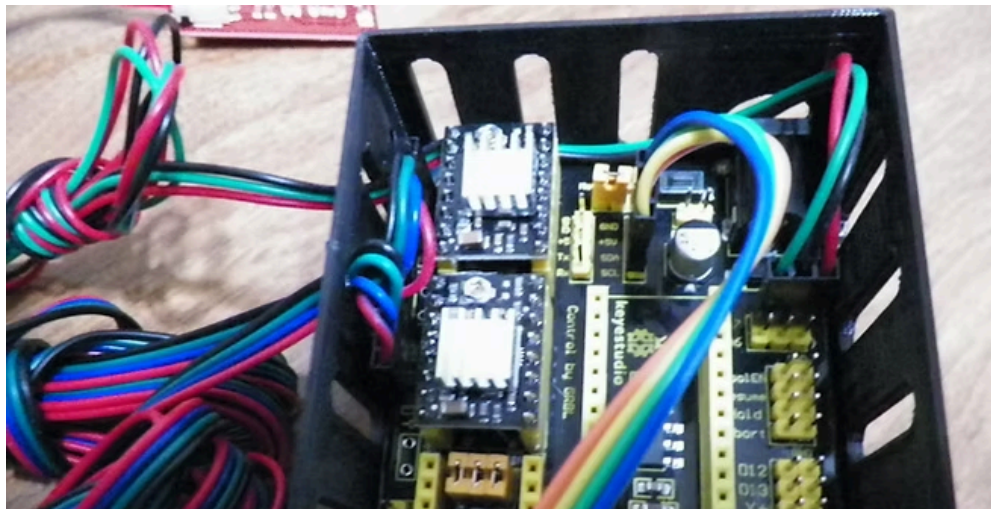
Catz 4 Catz
Mar 21



Hello Ace6879,
Did you swap drivers on the board and so did the problem fallow ?
Thank You



drivers iinstalled pic 1



Ace6879
Mar 21



Have you ever had an issue where only one motor will work even tho both are connected? I've swapped things around and used the test code also. I ordered another board just incase that's the issue.

Show more replies



Ace6879
Mar 31



Replying to [Ace6879](#)



Ace6879
Mar 31



Replying to [Ace6879](#)

If I switch the motor drivers, then the other motor doesn't work right



Liam Pulido
Mar 31



Replying to [Ace6879](#)

If you swap them and then the other one doesn't work, then it is definitely a fried driver. The video also looks like the driver is fried. Try swapping it in for the extra driver that would have been used for the Z-Axis. If that doesn't work, then replace it with the ones you ordered off ali (note: for the stepper drivers you don't have to buy keystudio ones. You can buy the extremely cheap ones). If even those don't work, then we can modify the code to use the Z-Axis driver slot instead of the X. Other than that, it looks like it is working fine!

Edited



Catz 4 Catz
Mar 18



Hello Liam,
Here are a couple of pictures.
Thank You Again For All Your Help. 🐱🐱



3D Chameleon Programed with CRC V4 Keystudio Board



Liam Pulido
Mar 18



Replying to [Catz 4 Catz](#)

That is great! It looks like everything is working now. The last thing to do is reassembling the chameleon unit. if you click the switch, something different should appear on the Oled and the motors should move! I am pretty sure that all the hard parts with programming the board are over! Hope this project turns out well for you!



Catz 4 Catz
Mar 18



Hello Liam,
Well to answer your question all 3 have been programed and all 3 screens say (3D Chameleon Mk4, Ready)
Thank You 🐱🐱



Catz 4 Catz
Mar 18



Hello Liam.
I never did ask which printer do you have your chameleon installed on ?
Thank You



Liam Pulido
Mar 18



Replying to [Catz 4 Catz](#)

I never got it set up on my printer. I have the Ankermake M5C (garbage compared to more recent printers, is completely closed source, and you can't have custom firmware.) which was a pain to try and set up on because:

- a) I cannot send direct gcode commands to it
- b) it's closed source so it's quite hard to design parts for it, like a filament cutter.
- c) I had some problems with my 3Dclippy, and I don't have the time to troubleshoot

In the end, I sold my unit to @ace6879 who probably will have an easier time than I did since he has an open-source printer that supports direct gcode commands. I am thinking of getting an Elegoo Centauri (not the carbon one) because it looks like a good machine with a good price, and it will have its own AMS clone (which will be revealed around June). Plus, even if their AMS clone doesn't get the best reviews, it has all the nifty features that are required for setting up a chameleon. Side question: Is the CNC Mainboard working?



Catz 4 Catz
Mar 18



Hello Liam,
Haven't set it up yet need to figure this gcode thing out I use Cura have never used Prusa I guess that Chris is using in his Chameleon Part 3
Thank You 🐱🐱



Liam Pulido
Mar 18



Replying to [Catz 4 Catz](#)
Well at least the hardware is working :)



Catz 4 Catz
Mar 18



Hello Liam,
Thank you so very much for all your help. 🐱🐱🐱🐱🐱🐱



Liam Pulido
Mar 18



Replying to [Catz 4 Catz](#)
Is it working?



Catz 4 Catz
Mar 18



Hello Liam,
You never did say is this the correct program ([SelectorFirmwareMk4.ino](#)) ?
Thank You



Liam Pulido
Mar 18



Replying to [Catz 4 Catz](#)

Yes, that is it, but you have the same pinout as me, you can simply copy this into Arduino IDE and skip the code modification:



ChameleonFirmwareCNCv4.txt
Download TXT • 29KB



Note: you cannot directly upload this into Arduino IDE, since it isn't a .ino file, and the website will not let me upload a .ino file, so you must copy and paste it.

Edited



Catz 4 Catz
Mar 17



Hello Liam,

You are the man works Great !!!!!!! Thank You Thank You 🐱🐱🐱🐱🐱🐱🐱🐱🐱

Thank You



Catz 4 Catz
Mar 17



Hello Liam,

Which Program ([SelectorFirmwareMk4.ino](#)) ?

Thank You



Liam Pulido
Mar 17



Replying to [Catz 4 Catz](#)

Try this:

```
#define EN      8    // stepper motor enable, low level effective
#define X_DIR   5    //X axis, stepper motor direction control
#define Y_DIR   6    //y axis, stepper motor direction control
#define Z_DIR   7    //zaxis, stepper motor direction control
#define X_STP   2    //x axis, stepper motor control
#define Y_STP   3    //y axis, stepper motor control
#define Z_STP   4    //z axis, stepper motor control
/*
// Function: step -control the direction and number of steps of the stepper motor
// Parameter: dir -direction control, dirPin corresponds to DIR pin, stepperPin corresponds
to

step pin, steps is the number of steps.
// no return value
*/
void step(boolean dir, byte dirPin, byte stepperPin, int steps)
{
```



```

digitalWrite(dirPin, dir);
delay(50);
for (int i = 0; i < steps; i++) {
  digitalWrite(stepperPin, HIGH);
  delayMicroseconds(800);
  digitalWrite(stepperPin, LOW);
  delayMicroseconds(800);
}
}

void setup()// set the IO pins for the stepper motors as output
pinMode(X_DIR, OUTPUT); pinMode(X_STP, OUTPUT);
pinMode(Y_DIR, OUTPUT); pinMode(Y_STP, OUTPUT);
pinMode(Z_DIR, OUTPUT); pinMode(Z_STP, OUTPUT);
pinMode(EN, OUTPUT);
digitalWrite(EN, LOW);
}

void loop(){
  step(false, X_DIR, X_STP, 3200); // x axis motor rotates CCW for 1 circle, as in 3200 steps
  step(false, Y_DIR, Y_STP, 3200); // y axis motor rotates CCW for 1 circle, as in 3200 steps
  step(false, Z_DIR, Z_STP, 3200); // z axis motor rotates CCW for 1 circle, as in 3200 steps
  delay(1000);
  step(true, X_DIR, X_STP, 3200); // X axis motor rotates CW for 1 circle, as in 3200 steps
  step(true, Y_DIR, Y_STP, 3200); // y axis motor rotates CW for 1 circle, as in 3200 steps
  step(true, Z_DIR, Z_STP, 3200); // z axis motor rotates CW for 1 circle, as in 3200 steps
  delay(1000);
}

```



Liam Pulido
Mar 17



Replying to [Liam Pulido](#)

This should work perfectly. The test code I sent you first had a fatal flaw. The stepping amount was set to 200, when it should have been 3200



Catz 4 Catz
Mar 17



Hello Liam,
What ever you did now it moves a 1/4 of a turn but not a full circle I would say your on the rite track.
Thank You



Liam Pulido
Mar 17



Replying to [Catz 4 Catz](#)

I found the problem! The testing code is wrong! Try using the testing code from here:
<https://www.3dchameleon.com/forum/main/comment/92bf7dfd-67ab-45d8-8d23-86de16afa65e?postId=6770a14a797c5c8b5ad99781>

and the problem should be solved!



Catz 4 Catz
Mar 17



Hello Liam

Keyestudio basically told me to pound sand because I didn't get the boards from them they were all ordered at the same time from Aliexpress and these clowns don't return e-mails. Did you load the code on yours? Don't know what to do seems like a waste of OUR TIME and money as I said I am really burned about this sorry. My guess is that the code is the problem.

According to other post other people supposedly have used this board why is it your the only one willing to help? Even Bill said this was an easier choice I would have though he would have chimed in by now. I can't believe all 3 boards and 8 nema17 steppers are be bad

Thank You



Liam Pulido
Mar 17



Replying to [Catz 4 Catz](#)

I do not think that the steppers are bad. I think its the code. I think that if the boards do the same thing anyway, we should just go ahead and try with the chameleon firmware. Also I'm just the main person who helps with the CNC shields because:

- 1) Bill is busy and he has a stock of mainboards for personal use, so he doesn't need a CNC Shield.
- 2) Other people probably don't check the forums as much as I do.

I do have a modified code I would like you to try and record the results.



Liam Pulido
Mar 17



Replying to [Catz 4 Catz](#)

Code:

```
#define EN      8    // stepper motor enable, low level effective
#define X_DIR   5    //X axis, stepper motor direction control
#define Y_DIR   6    //y axis, stepper motor direction control
#define Z_DIR   7    //zaxis, stepper motor direction control
#define X_STP   2    //x axis, stepper motor control
#define Y_STP   3    //y axis, stepper motor control
#define Z_STP   4    //z axis, stepper motor control
/*
// Function: step -control the direction and number of steps of the stepper motor
// Parameter: dir -direction control, dirPin corresponds to DIR pin, stepperPin corresponds
to

step pin, steps is the number of steps.
// no return value
*/
void step(boolean dir, byte dirPin, byte stepperPin, int steps)
{
    digitalWrite(dirPin, dir);
    delay(50);
    for (int i = 0; i < steps; i++) {
        digitalWrite(stepperPin, HIGH);
        delayMicroseconds(800);
```

```
digitalWrite(stepperPin, LOW);
delayMicroseconds(800);
}
}

void setup(){// set the IO pins for the stepper motors as output
pinMode(X_DIR, OUTPUT); pinMode(X_STP, OUTPUT);
pinMode(Y_DIR, OUTPUT); pinMode(Y_STP, OUTPUT);
pinMode(Z_DIR, OUTPUT); pinMode(Z_STP, OUTPUT);
pinMode(EN, OUTPUT);
digitalWrite(EN, LOW);
}

void loop(){
  step(false, X_DIR, X_STP, 800); // x axis motor rotates CCW for 1 circle, as in 800 steps
  step(false, Y_DIR, Y_STP, 800); // y axis motor rotates CCW for 1 circle, as in 800 steps
  step(false, Z_DIR, Z_STP, 800); // z axis motor rotates CCW for 1 circle, as in 800 steps
  delay(1000);
  step(true, X_DIR, X_STP, 800); // X axis motor rotates CW for 1 circle, as in 800 steps
  step(true, Y_DIR, Y_STP, 800); // y axis motor rotates CW for 1 circle, as in 800 steps
  step(true, Z_DIR, Z_STP, 800); // z axis motor rotates CW for 1 circle, as in 800 steps
  delay(1000);
}
```



Catz 4 Catz
Mar 17



Guess you have given up at this point :(By the way all 3 boards are doing the something this has been a waste of my time and your's so much for this being the greatest board sorry just a little burned.
Thanks



Liam Pulido
Mar 17



Replying to [Catz 4 Catz](#)

Did you order all the boards at the same time?

If you did, keyestudio may have sent you a faulty batch. You should complain to them and ask for a new board. If that board doesn't work, then we know its the code thats the problem.



Liam Pulido
Mar 17



Replying to [Liam Pulido](#)

Also, since the motors do spin correctly, maybe we should just try uploading the 3D chameleon firmware and see if it just works for some reason.



Catz 4 Catz
Mar 17



Hello Liam,

I am at a loss I tried another set of nema17 steppers they do the samething. I tried 6 diffrent 12 power supplys (2a to 5a) samething nothing has changed I even reinstalled arduino ide to 1.8.18

 Liam Pulido
Mar 17

Replying to [Catz 4 Catz](#)

I am doing some reasearch on what may be causing this. It almost looks like its not going the correct amount of steps. I will have some modified code for you to try in a bit.

 Liam Pulido
Mar 17

Replying to [Catz 4 Catz](#)

Here a a few suggestions:

Go to <https://forum.arduino.cc/> and make a post there. This is the arduino formus. There are people who are smarter than me and can probably be more helpful. Make sure to include the video of the problem, the screenshot of the code, and the fixes that you have already tried.

Message keystudio. Their board is the thing that is having problems, so just ask them about it.



 Catz 4 Catz
Mar 16

Hello Liam,

The motors are good there the back up's for my enders 5's both sets do the same thing. They work fine in the ender 5's even the ones I ordered from amazon do the same thing as I said I have 3 set's of everything as I have 3 ender 5.

Thank You

 Catz 4 Catz
Mar 16

Hello Liam,
Here you go.

Thank You

```
1 #define EN 8 // stepper motor enable, low level effective
2 #define X_DIR 5 //X axis, stepper motor direction control
3 #define Y_DIR 6 //y axis, stepper motor direction control
4 #define Z_DIR 7 //zaxis, stepper motor direction control
5 #define X_STP 2 //X axis, stepper motor control
6 #define Y_STP 3 //y axis, stepper motor control
7 #define Z_STP 4 //z axis, stepper motor control
8 /*
9 // Function: step -control the direction and number of steps of the stepper motor
10 // Parameter: dir -direction control, dirPin corresponds to DIR pin, stepperPin corresponds to
11
12
13 step pin, steps is the number of steps.
14 // no return value
15 */
16 void step(boolean dir, byte dirPin, byte stepperPin, int steps)
17 {
18     digitalWrite(dirPin, dir);
19     delay(50);
20     for (int i = 0; i < steps; i++) {
21         digitalWrite(stepperPin, HIGH);
22         delayMicroseconds(800);
23         digitalWrite(stepperPin, LOW);
24         delayMicroseconds(800);
25     }
26 }
27 void setup(){// set the IO pins for the stepper motors as output
28     pinMode(X_DIR, OUTPUT); pinMode(X_STP, OUTPUT);
29     pinMode(Y_DIR, OUTPUT); pinMode(Y_STP, OUTPUT);
30     pinMode(Z_DIR, OUTPUT); pinMode(Z_STP, OUTPUT);
31     pinMode(EN, OUTPUT);
32     digitalWrite(EN, LOW);
33 }
34 void loop(){
35     step(false, X_DIR, X_STP, 200); // x axis motor rotates CCW for 1 circle, as in 200 steps
36     step(false, Y_DIR, Y_STP, 200); // y axis motor rotates CCW for 1 circle, as in 200 steps
37     step(false, Z_DIR, Z_STP, 200); // z axis motor rotates CCW for 1 circle, as in 200 steps
38     delay(1000);
39     step(true, X_DIR, X_STP, 200); // X axis motor rotates CW for 1 circle, as in 200 steps
40     step(true, Y_DIR, Y_STP, 200); // y axis motor rotates CW for 1 circle, as in 200 steps
41     step(true, Z_DIR, Z_STP, 200); // z axis motor rotates CW for 1 circle, as in 200 steps
42     delay(1000);
43 }
44
```



Catz 4 Catz
Mar 16

:

Hello Liam,
Here is the video
Thank You



Liam Pulido
Mar 16

⋮

Replying to [Catz 4 Catz](#)

Strange. The motors are rotating in the correct directions but aren't making the full turn. This probably won't give me more than I already know, but could you take a screenshot of your testing code? Also, try plugging these into one of your ender 5's (just unplug a motor and plug it into one of these) and move the axis you plugged it into. This is to see if the motors themselves are faulty. It also may be worth trying to test another set of stepper motors.



Catz 4 Catz
Mar 16

⋮

Hello Liam,

HELP ! Ok this is where I am at plug the nano in to program power led and blinking L led ran program just the power led is on now. Plug shield into 12v screen doesnt work,the setppers both X,Y move about a 1/8 of a turn CCW and CW directions say 1 full turn (circle) ? Don't know what the deal is with the screen wires are pin for pin ? I screwed something up I guess.

Thank You 🐱🐱🐱



Liam Pulido
Mar 16

⋮

Replying to [Catz 4 Catz](#)

While using the test code, the screen will not power on, which means that part is working normally. The test code will only affect the motors. Could you send a video of the motors spinning? I have never had that happen, so I must see what goes wrong before I can help. Also, it may be worth taking the screen out of the lid temporarily so I can see the pin locations.

[Edited](#)



Catz 4 Catz
Mar 15

⋮

Hello Liam,

This may sound dumb but I really don't get the program part ? The link you sent 2 areas top and bottom which one do you cut and paste for test ? And what do you program to the board ? When it comes to putting things together I am good at that but programing not so much.

Thank You 🐱🐱



Liam Pulido
Mar 15

⋮

Replying to [Catz 4 Catz](#)

```
#define EN      8    // stepper motor enable, low level effective
#define X_DIR   5     //X axis, stepper motor direction control
#define Y_DIR   6     //y axis, stepper motor direction control
#define Z_DIR   7     //zaxis, stepper motor direction control
```



```

#define X_STP 2 //x axis, stepper motor control
#define Y_STP 3 //y axis, stepper motor control
#define Z_STP 4 //z axis, stepper motor control
/*
// Function: step -control the direction and number of steps of the stepper motor
// Parameter: dir -direction control, dirPin corresponds to DIR pin, stepperPin corresponds
to

step pin, steps is the number of steps.
// no return value
*/
void step(boolean dir, byte dirPin, byte stepperPin, int steps)
{
    digitalWrite(dirPin, dir);
    delay(50);
    for (int i = 0; i < steps; i++) {
        digitalWrite(stepperPin, HIGH);
        delayMicroseconds(800);
        digitalWrite(stepperPin, LOW);
        delayMicroseconds(800);
    }
}

void setup(){// set the IO pins for the stepper motors as output
    pinMode(X_DIR, OUTPUT); pinMode(X_STP, OUTPUT);
    pinMode(Y_DIR, OUTPUT); pinMode(Y_STP, OUTPUT);
    pinMode(Z_DIR, OUTPUT); pinMode(Z_STP, OUTPUT);
    pinMode(EN, OUTPUT);
    digitalWrite(EN, LOW);
}

void loop(){
    step(false, X_DIR, X_STP, 200); // x axis motor rotates CCW for 1 circle, as in 200 steps
    step(false, Y_DIR, Y_STP, 200); // y axis motor rotates CCW for 1 circle, as in 200 steps
    step(false, Z_DIR, Z_STP, 200); // z axis motor rotates CCW for 1 circle, as in 200 steps
    delay(1000);
    step(true, X_DIR, X_STP, 200); // X axis motor rotates CW for 1 circle, as in 200 steps
    step(true, Y_DIR, Y_STP, 200); // y axis motor rotates CW for 1 circle, as in 200 steps
    step(true, Z_DIR, Z_STP, 200); // z axis motor rotates CW for 1 circle, as in 200 steps
    delay(1000);
}

```



Liam Pulido
Mar 15



Replying to [Liam Pulido](#)

This is the code (the other reply). You will copy and paste this into Arduino IDE, and then upload it to the Arduino nano (inserted into the CNC Shield). Then use my previous comment to see which direction each motor should spin.

<https://www.3dchameleon.com/forum/main/comment/9823011a-16b3-4296-a865-fab838550bfa?postId=6770a14a797c5c8b5ad99781>

Edited



Catz 4 Catz
Mar 14



Hello Liam,

This is the Y adapter I am using it has a nice smooth flow and it has a fifth spot if I only want to use a single filament. Thank you for your compliment on my setup so far. 🐱🐱



My Y Adapter



Liam Pulido
Mar 15



Replying to [Catz 4 Catz](#)

Great! After the board has been programmed, you will be done with the mainboard setup, and you will be able to continue with the chameleon setup like normal. Unfortunately, this is where my help will end, as I am not good with the gcode configuration and the other bits.

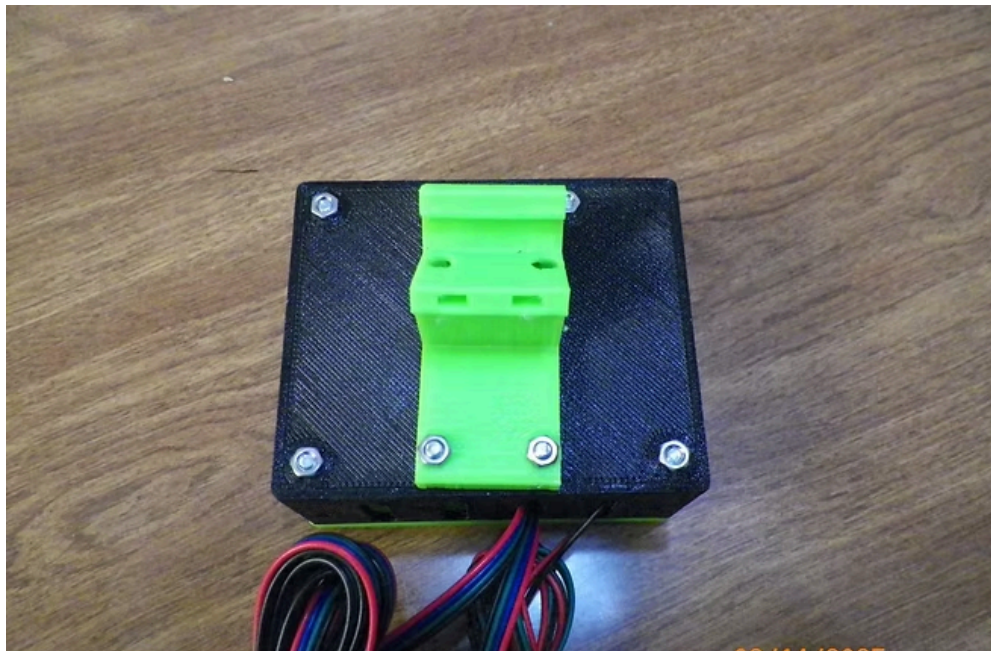
Chris Riley has a very helpful 3 part series that I originally followed to get my hardware setup. Part one can be found here: <https://www.youtube.com/watch?v=NUYJseo78tI>



Catz 4 Catz
Mar 14



Hello Liam,



Liam Pulido
Mar 15



Replying to [Catz 4 Catz](#)

Funnily enough, I didn't even set in any M3 nuts before screwing in the M3x10 bolts. I made the hole sizes be small enough that the bolts could self-thread, and sit in the plastic (which isn't the best) but works for now. I probably should have done what you did, but I just didn't have any M3 nuts on hand. I will probably add in holes to the back of the model so next time, you won't have to drill.

Edited



Catz 4 Catz
Mar 14



Hello Liam,



Liam Pulido
Mar 14



Replying to [Catz 4 Catz](#)

This setup is looking quite nice! You should probably mount the end stop switch at the corner by the mainboard box, to avoid cables going everywhere. One last upgrade that is definitely worth doing is the 3Dclippy. It removes the need for tip shaping (which can be an absolute pain). I don't know if there are any mounts made directly for the ender-5 series, but the extruder plates look quite similar to the ender-3, so it probably isn't that hard to adapt (though it may interfere with your current direct drive setup).

A few models that may or may not be helpful

<https://www.printables.com/model/891355-3d-chameleon-auto-clippy-mount-micro-swiss-ng>

<https://www.printables.com/model/473548-3dchameleon-ender-5-extruder-mount-with-ptfe>

<https://www.tinkercad.com/things/6Lnp2FgNGZg-digital3dclippyender3v2>

<https://www.tinkercad.com/things/bw2TejY3FwY-digital3dclippyender3neo>

As for the 4-Y PTFE adapter (which hasn't even been mentioned yet), I don't recommend printing the 3DChameleon one. Instead, get a smooth Bambu Lab one like this:

https://www.aliexpress.us/item/3256806861554523.html?spm=a2g0o.productlist.main.5.358c5da3LdK0H&algo_pvid=ec376282-fd63-4b91-b060-b8463a534a89&algo_exp_id=ec376282-fd63-4b91-b060-b8463a534a89-2&pdp_ext_f=%7B%22order%22%3A%22349%22%2C%22eval%22%3A%221%22%7D&pdp_npi=4%40dis%21USD%216.98%216.98%21%21%2150.34%2150.34%21%402103245417419801483232822e2fb0%2112000039217455870%21sea%21US%210%21ABX&curPageLogUId=vh1QkZOOct60z&utparam-url=scene%3Asearch%7Cquery_from%3A

Edited



Catz 4 Catz
Mar 14



Hello Liam

Here are pictures of how your CRC V4 box is mounted on my Ender 5 Plus. I drill the back of your box and installed hook with 4 3m x 10 screws and it just hangs in the 2020 rail. 🐱🐱

Thank You



Catz 4 Catz
Mar 14



Hello Liam,
Sorry 2 Ender 5 Pro and 1 End 5 Plus 🐱🐱
Thank You



Catz 4 Catz
Mar 14



Hello Liam,
Looks Great ! 🐱🐱🐱



Catz 4 Catz
Mar 14



Hello Liam,
2 Ender Pro and 1 Ender Plus. I have made 3 of these chameleon units. 🐱🐱
Thank You



Liam Pulido
Mar 14



Replying to [Catz 4 Catz](#)
Are these the correct printer models?
<https://www.creality.com/products/ender-3-pro-3d-printer>
<https://www.creality.com/products/creality-ender-3-s1-plus-3d-printer>

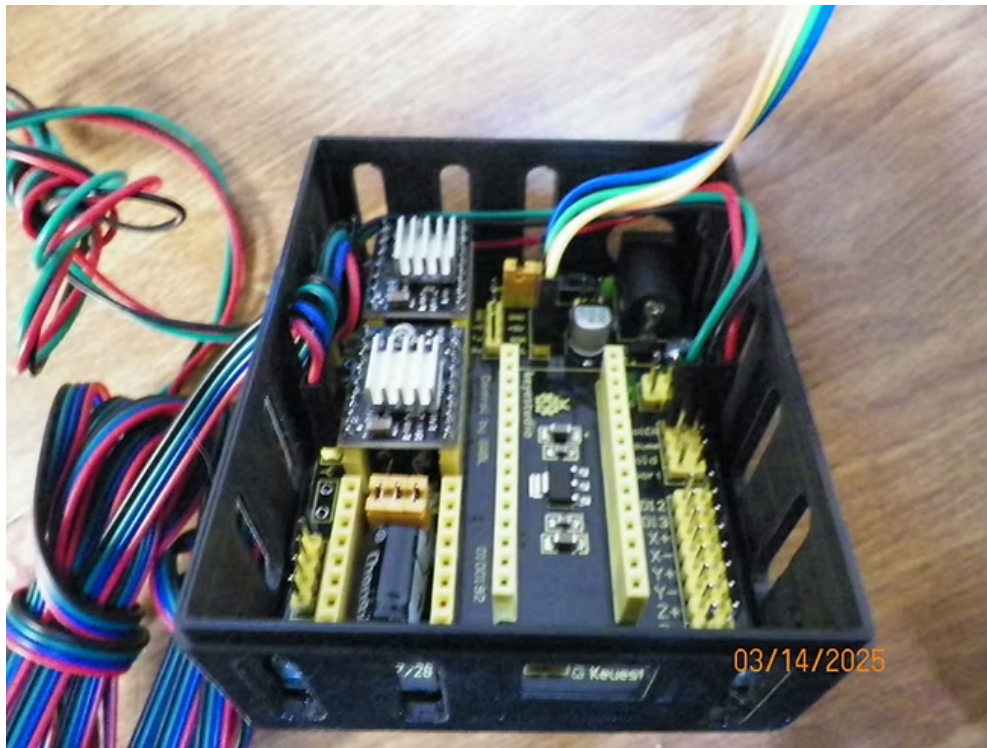
You didn't specify which ender model (3, 5, 3 v3, etc.) so just wondering, as some printers have some more developed upgrades, and would be easier to make a mainboard box mount for



Catz 4 Catz
Mar 14



Hello Liam,
Here is the picture you requested. 🐱🐱
Thank You



Liam Pulido
Mar 14



Replying to [Catz 4 Catz](#)
Alright, thats looking better. That was a needed adjustment, as it wouldn't work with the modified code otherwise.



Catz 4 Catz
Mar 14



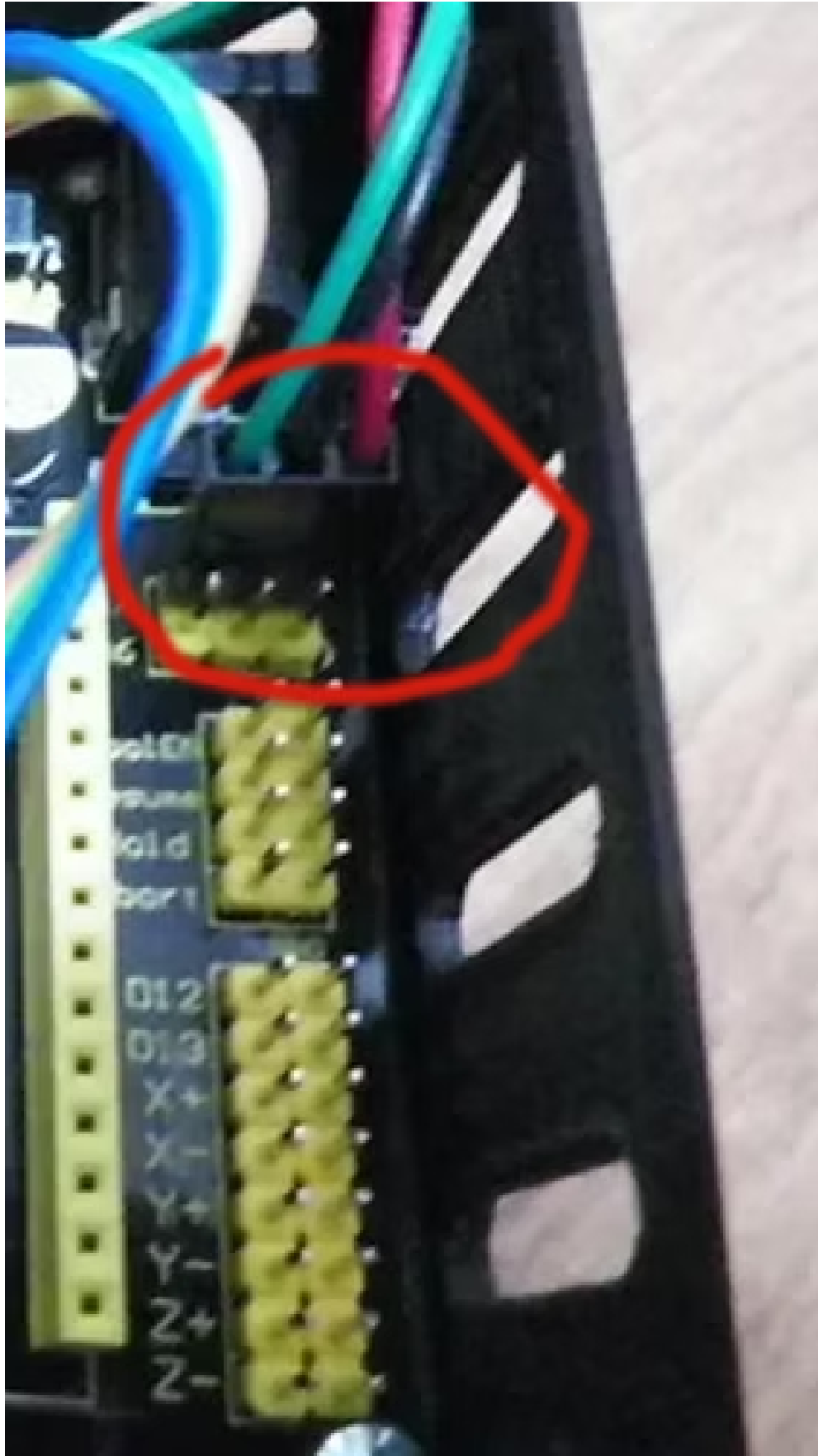
Hello Liam,
Thank you for pointing that out 🐱🐱



Liam Pulido
Mar 14



Just a note, you made a minor mistake in your wiring.
You put the limit switch end connector here:



Your current pinout

While it should be here:



The red cable end should be sticking off the edge, not connected to anything. The black cable end should be connected to the rightmost side (assuming you are looking at it the same way as in the picture) and the green should be connected to the leftmost circled pin.

Just something to watch out for, as that's an easy way to fry the CNC Shield.



Catz 4 Catz
Mar 14



Hello Liam,
Sure not a problem. 🐱🐱



Liam Pulido
Mar 14



Replying to [Catz 4 Catz](#)
Okay, it is now posted!

Link: <https://makerworld.com/en/models/1207140-3d-chameleon-cnc-shield-v4-case#profileId-1221194>



Liam Pulido
Mar 14



Replying to [Catz 4 Catz](#)
Also, quick question: what printer are you setting this up on? some are easier to setup on than others.



Catz 4 Catz
Mar 13



Hello Liam,
Sure not a problem 🐱🐱
Thank You



Catz 4 Catz
Mar 12



Hello Liam,
Again thank you for the help 🐱🐱



Liam Pulido
Mar 13



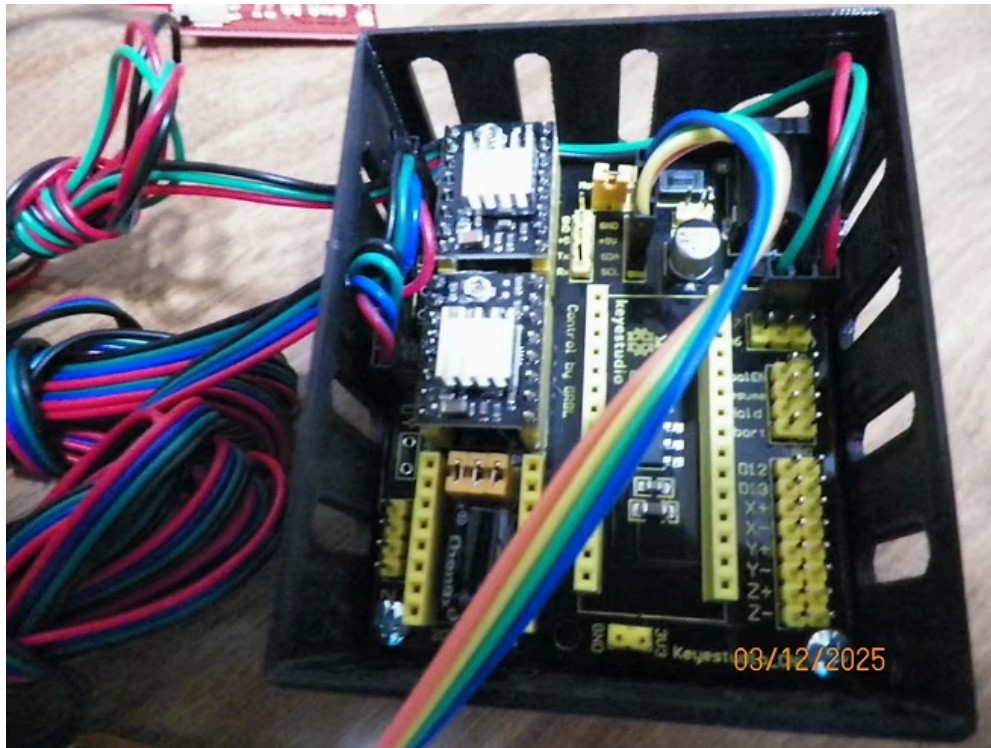
Replying to [Catz 4 Catz](#)
Quick question: Is it ok if I use your photos of my CNC Shield box so I can post it on makeworld?



Catz 4 Catz
Mar 12



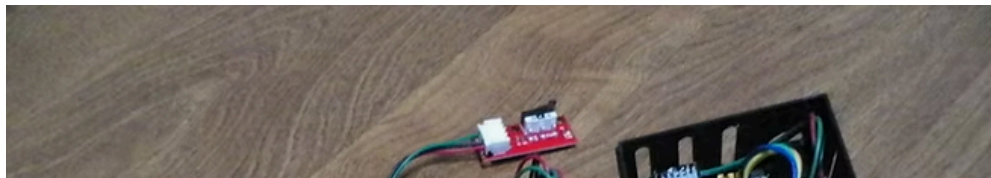
Hello Liam,



Catz 4 Catz
Mar 12



Hello Liam,



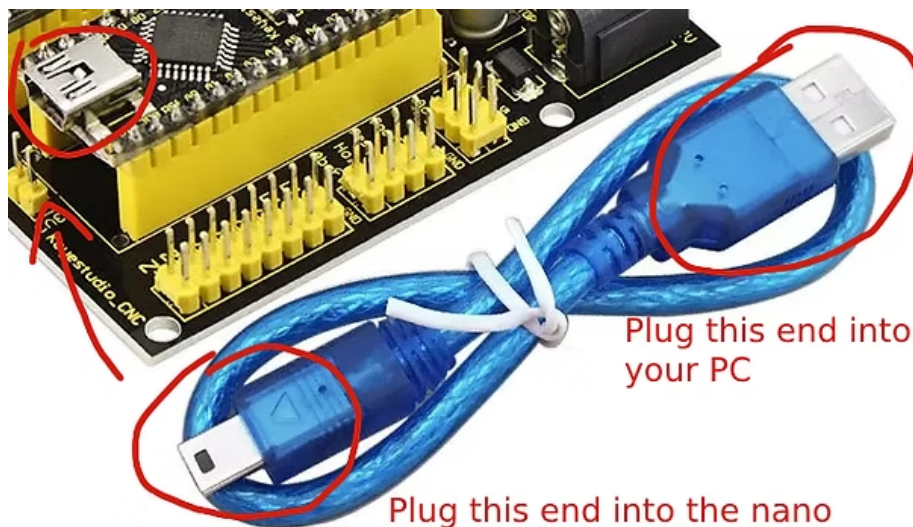
Liam Pulido
Mar 12



Replying to [Catz 4 Catz](#)

Okay that is looking great! You are almost there! I do see that my CNC Shield box is getting used ;)

This is the time where you will test the motor directions. I can see that the Arduino nano itself is not inserted into the board. Before continuing, proceed to do that. My video of inserting stepper drivers also shows putting in the nano. After that, you will need to plug the small end of the blue cable into the nano boards USB port, then plug the full size USB end into your pc's USB port.



Plug the small end into the nano, and the large end into your PC

After this, the Arduino program should recognize the nano board, and it will let you upload code to it. Also, before giving power to the main board, make sure that the OLED screen has the correct pinout. I showed how to do this on one of my comments on Daniel's post. After this, make sure that the motors and arduino nano are connected to the CNC Shield. Then, you can plug in your 12v power supply. Now onto the motor testing. First, delete all code already in the code box in Arduino IDE, then copy the code from this post: <https://www.3dchameleon.com/forum/main/comment/92bf7dfd-67ab-45d8-8d23-86de16afa65e?postId=6770a14a797c5c8b5ad99781>

Using the notes in the code, you should be able to see if each motor turns the correct direction when they are supposed too.

Here is the order and directions that the motors should spin while using the test code:

First, the X-Motor (in our case the extruder motor) should rotate counterclockwise (to the left) for one circle.

Then, the Y-Motor (which for you would be the selector) should rotate counterclockwise for one circle.

After, the Z-Motor would do the same, but we only have 2 motors in the chameleon, so ignore the Z rotations.

Then after that, these rotations will happen:

The X-Motor should rotate clockwise (to the right) for one circle.

Then, the Y-Motor (which for you would be the selector) will rotate clockwise for one circle.

If they turn the wrong direction, flip the cable ends that go into the CNC Shield 180 degrees then try again.

Edited



Catz 4 Catz
Mar 12



Hello Liam,
This is where I am at now thanks to your help.
Thank You





Mar 12



Hello Liam,

I haven't figure out how to program this thing , I downloaded the Arduino program and got lost as you can tell I am new to all this as the real 3d chameleon board would have been pre programed.

Thank You



Liam Pulido

Mar 12



Replying to [Catz 4 Catz](#)

Have you followed the guide in my other reply? I would first recommend testing for motor directions. You will need to have the stepper drivers and nano installed into the CNC Shield. Then, you will need to connect the mini USB cable into the nano and plug the other end into your pc. The computer should recognize the nano board, and you should be able to send code to the board.



Catz 4 Catz

Mar 12



Hello Liam,

Sorry I ask so many questions.

Thank You



Liam Pulido

Mar 12



Replying to [Catz 4 Catz](#)

Its fine! I am just here to help. After all, I sold my chameleon unit, but I still love the 3D printing community, and the idea of an open source MMU that works on any printer. I may build another unit for an Elegoo Centauri if the official MMU is too expensive.



Liam Pulido

Mar 12



Replying to [Catz 4 Catz](#)

Also, how is the setup turning out? Is it working okay?



Catz 4 Catz

Mar 12



Hello Liam,

I ment which one is selector and which one is extractor ?

Thank You



Comment

Liam Pulido
Mar 12

Replying to [Catz 4 Catz](#)

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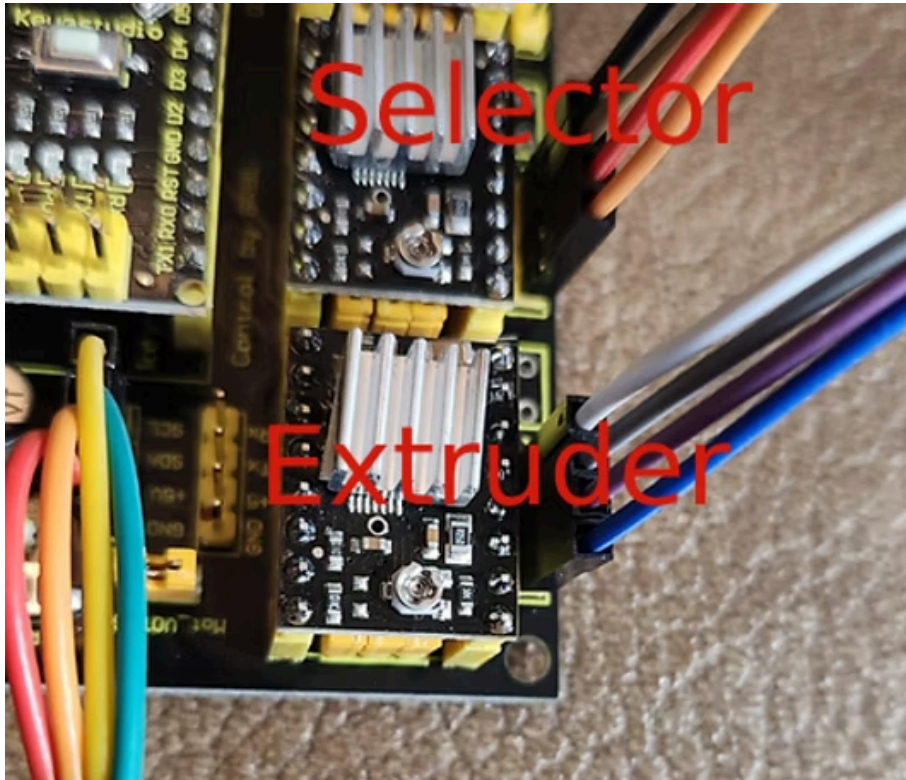
194 comments

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Setting up Ender 3 S1

First time set up.

Firmware changes for Mk4.1
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The top stepper is the Y-Axis (selector) and the bottom is the X-Axis (extruder)

For the latest information on product availability, be sure to follow us on twitter! @3DChameleon

For more installation videos, be sure to follow us on Youtube!

Contact us: info@3dchameleon.com



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Each stepper driver in the kit is the same. Just follow the simple board assembly video I put in my reply on the other post.



Tyeth Gundry
Jan 08



I had slightly different pin requirements, I'm using the v4 nano (and a v3 uno), except it's the HW-702 v0.0.0, and in addition to fixing the microstep jumpers to 5V instead of ground, I had to alter the step/direction pins to accommodate the differences according to this info:

<https://www.instructables.com/Fix-Cloned-Arduino-NANO-CNC-Shield/#:~:text=Lines%2047%2C48%20%26%2049-,need%20replacing%20with,-%3A>

My changes are here (also includes optional serial prints for display-less diagnostics)

<https://github.com/3DChameleon/3DChameleonMk4/compare/main...tyeth:3DChameleonMk4:nano-cnc-shield>

[Edited](#)

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